UNIT 2

History of computers

When you read the following text, you will probably meet words and expressions that are new to you. First try to understand their meaning from the context – read the same passage a few times. When you have read the whole text, check new words in a dictionary. Most of the words in **bold** *typeface are explained in the Glossary at the end of this book.*

[1] Let us take a look at the history of the computers that we know today. The very first calculating device used was the ten fingers of a man's hands. This, in fact, is why today we still count in tens and multiples of tens. Then the **abacus** was invented, a bead frame in which the beads are moved from left to right. People went on using some form of abacus well into the 16th century, and it is still being used in some parts of the world because it can be understood without knowing how to read.

[2] During the 17th and 18th centuries many people tried to find easy ways of calculating. J. Napier, a Scotsman, devised a mechanical way of multiplying and dividing, which is how the modern **slide rule** works. Henry Briggs used Napier's ideas to produce **logarithm tables** which all mathematicians use today. **Calculus,** another branch of mathematics, was independently invented by both Sir Isaac Newton, an Englishman, and Leibnitz, a German mathematician.

[3] The first real calculating machine appeared in 1820 as the result of several people's experiments. This type of machine, which saves a great deal of time and reduces the possibility of making mistakes, depends on a series of ten-toothed gear wheels. In 1830 Charles Babbage, an Englishman, designed a machine that was called 'The Analytical Engine'. This machine, which Babbage showed at the Paris Exhibition in 1855, was an attempt to cut out the human being altogether, except for providing the machine with the necessary facts about the problem to be solved. He never finished this work, but many of his ideas were the basis for building today's computers.

[4] In 1930, the first **analog** computer was built by an American named Vannevar Bush. This device was used in World War II to help aim guns. Mark I, the name given to the first **digital** computer, was completed in 1944. The men responsible for this invention were Professor Howard Aiken and some people from IBM. This was the first

machine that could figure out long lists of mathematical problems, all at a very fast rate. In 1946 two engineers at the University of Pennsylvania, J. Eckert and J. Mauchly, built the first digital computer using parts called **vacuum tubes**. They named their new invention ENIAC. Another important advancement in computers came in 1947, when John von Newmann developed the idea of keeping instructions for the computer inside the computer's memory.

[5] The first generation of computers, which used vacuum tubes, came out in 1950. Univac I is an example of these computers which could perform thousands of calculations per second. In 1960, the second generation of computers was developed and these could perform work ten times faster than their predecessors. The reason for this extra speed was the use of **transistors** instead of vacuum tubes. Second-generation computers were smaller, faster and more dependable than first-generation computers. The third-generation computers appeared on the market in 1965. These computers could do a million calculations a second, which is 1000 times as many as first-generation computers. Unlike second-generation computers, these are controlled by tiny integrated circuits and are consequently smaller and more dependable. Fourth-generation computers have now arrived, and the integrated circuits that are being developed have been greatly reduced in size. This is due to **microminiaturization**, which means that the circuits are much smaller than before; as many as 1000 tiny circuits now fit onto a single **chip**. A chip is a square or rectangular piece of silicon, usually from $\frac{1}{10}$ to $\frac{1}{4}$ inch, upon which several layers of an integrated circuit are etched or imprinted, after which the circuit is encapsulated in plastic, ceramic or metal. Fourth-generation computers are 50 times faster than third-generation computers and can complete approximately 1,000,000 instructions per second.

[6] At the rate computer technology is growing, today's computers might be obsolete by 1985 and most certainly by 1990. It has been said that if transport technology had developed as rapidly as computer technology, a trip across the Atlantic Ocean today would take a few seconds.

Exercises

1 Main idea

Which statement best expresses the main idea of the text? Why did you eliminate the other choices?

☐ 1. Computers, as we know them today, have gone through many changes.

☐ 2. Today's computer probably won't be around for long.

☐ 3. Computers have had a very short history.

2 Understanding the passage

Decide whether the following statements are true or false (T/F) by referring to the information in the text. Then make the necessary changes so that the false statements become true.

T	F	
☐	☐	1. The abacus and the fingers are two calculating devices still in use today.
☐	☐	2. The slide rule was invented hundreds of years ago.
☐	☐	3. During the early 1880s, many people worked on inventing a mechanical calculating machine.
☐	☐	4. Charles Babbage, an Englishman, could well be called the father of computers.
☐	☐	5. The first computer was invented and built in the USA.
☐	☐	6. Instructions used by computers have always been kept inside the computer's memory.
☐	☐	7. Using transistors instead of vacuum tubes did nothing to increase the speed at which calculations were done.
☐	☐	8. As computers evolved, their size decreased and their dependability increased.
☐	☐	9. Today's computers have more circuits than previous computers.
☐	☐	10. Computer technology has developed to a point from which new developments in the field will take a long time to come.

3 Locating information

Find the passages in the text where the following ideas are expressed. Give the line references.

.... 1. During the same period in history, logarithm tables and calculus were developed.

.... 2. It wasn't until the 19th century that a calculating machine was invented which tried to reduce manpower.

.... 3. Integrated circuitry has further changed computers.

.... 4. People used their fingers to count.

.... 5. The computers of the future may be quite different from those in use today.

.... 6. Today's computer circuits can be put on a chip.

.... 7. Then an instrument with beads was invented for counting before a mechanical way for multiplying and dividing was devised.

.... 8. Transistors replaced vacuum tubes.

4 Understanding words

Refer back to the text and find synonyms (i.e. words with a similar meaning) for the following words.

1. machine	(*l. 2*)	....
2. designed	(*l. 9*)	....
3. a lot	(*l. 16*)	....
4. errors	(*l. 17*)	....
5. solve	(*l. 30*)	....

Now refer back to the text and find antonyms (i.e. words with an opposite meaning) for the following words.

6. old	(*l. 10*)	....
7. a few	(*l. 16*)	....
8. to include	(*l. 21*)	....
9. contemporaries	(*l. 41*)	....
10. still in use	(*l. 60*)	....

5a Content review

Match the following words in column A with the statements in column B. The first one is done for you.

	A	B
c	1. abacus	a. instrument used for doing multiplication and division
☐	2. calculus	b. used in the first digital computers
☐	3. analog computer	c. an instrument used for counting
☐	4. digital computer	d. used in mathematics
☐	5. vacuum tubes	e. circuitry of fourth-generation computers
☐	6. transistors	f. invented by Americans in 1944
☐	7. chip	g. made computers smaller and faster
☐	8. microminiaturization	h. used to help aim guns
☐	9. slide rule	i. the reduction of circuitry onto a chip
☐	10. logarithm tables	j. a branch of mathematics